

Prototype-Guided Multi-Task Learning for HCC

Postoperative Complications Prediction

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Supplementary Materials:

Supplementary Text S1. Baseline Characteristics

Supplementary Table I summarizes the baseline characteristics of patients in the development cohort (n=500) and the evaluation cohort (n=126). The EHR dataset are systematically categorized into four groups: Basic Variables, Preoperative Factors, Intraoperative Factors, and Postoperative Factors. For statistical comparisons between the two cohorts, continuous variables were analyzed using either the independent t-test or the Mann-Whitney U test, depending on the data distribution. Categorical variables were evaluated using the Chi-square test or Fisher's exact test.

Supplementary Text S2. Performance of our model as an AI assistant for clinicians

To comprehensively validate the model's clinical utility across different types of complications, we extended our robustness and human-AI collaboration evaluations to pulmonary outcomes (Fig. S2). Consistent with the primary findings, the proposed framework demonstrated graceful degradation under simulated data sparsity, maintaining an AUC well above 0.70 for both pneumonia (PNA) and pleural effusion (PE) even when 40% of the structured data was missing (Fig. S2A). Furthermore, as shown in Fig. S2B, the experiment confirmed that integrating our model predictions as a second opinion noticeably improved clinicians' diagnostic precision for these systemic complications. AI-assisted clinicians achieved higher median

AUC and F1-scores compared to their independent baseline assessments, underscoring the framework's versatility as a reliable decision-support tool beyond liver-specific risks.

Supplementary Text S3. Analysis of Label Dependency and Task Correlation

To address concerns regarding potential shortcut learning due to the hierarchical overlap between CCI and individual complications, we performed a leave-one-out ablation analysis on the auxiliary tasks. As shown in Supplementary Table II, removing individual complication labels (PHLF, PNA, PE, or ASC) did not lead to a collapse in CCI prediction performance. Notably, even after removing specific complications, the model maintained higher or comparable performance relative to the STL baseline. These results indicate that the performance gains of our MTL framework stem from the comprehensive modeling of the patient's holistic postoperative state rather than mere label redundancy.

---Role---
 -You are a medical expert specializing in clinical research for hepatobiliary surgery. Your task is to extract and integrate all risk features related to predicting post-operative complications of HCC from complex [Admission Records].

---Goal---
 -Strictly control the total length between 120-150 characters. Describe facts objectively; avoid redundancy, diagnostic conclusions, or clinical advice. Specific values for indicators such as AFP and BP must be retained. Do not include specific dates or hospital names.

Key Extraction Points:
 - Liver Reserve: HBV history, AFP value, signs of cirrhosis/portal hypertension.
 - Tumor Status: Specific location, capsule, and hemorrhage status.
 - Systemic Risk Indicators: Smoking/alcohol history, hypertension, pulmonary inflammation.

Surgical Records Input Prompt

---Role---
 -You are a medical expert specializing in clinical research for hepatobiliary surgery. Your task is to extract and integrate all risk features related to predicting post-operative complications of HCC from complex [Surgical Records].

---Goal---
 -Strictly control the total length between 120-150 characters. Prohibit any subjective suggestions, diagnostic conclusions, or clinical guidance. Describe only the "patient's surgical course." Must retain hard indicators such as blood loss, tumor diameter, and incision length. Do not include specific dates or hospital names.

Key Extraction Points:
 - Surgical Strategy: Whether converted to open surgery.
 - Liver and Tumor Characteristics: Degree of cirrhosis, tumor size, and presence of rupture.
 - Core Technical Indicators: Resection scope, hepatic blood flow occlusion, and intraoperative blood loss.
 - Concomitant Procedures: Whether combined with cholecystectomy or adhesiolysis.

Figure S1. Prompt templates for extracting HCC risk features from admission and surgical records.

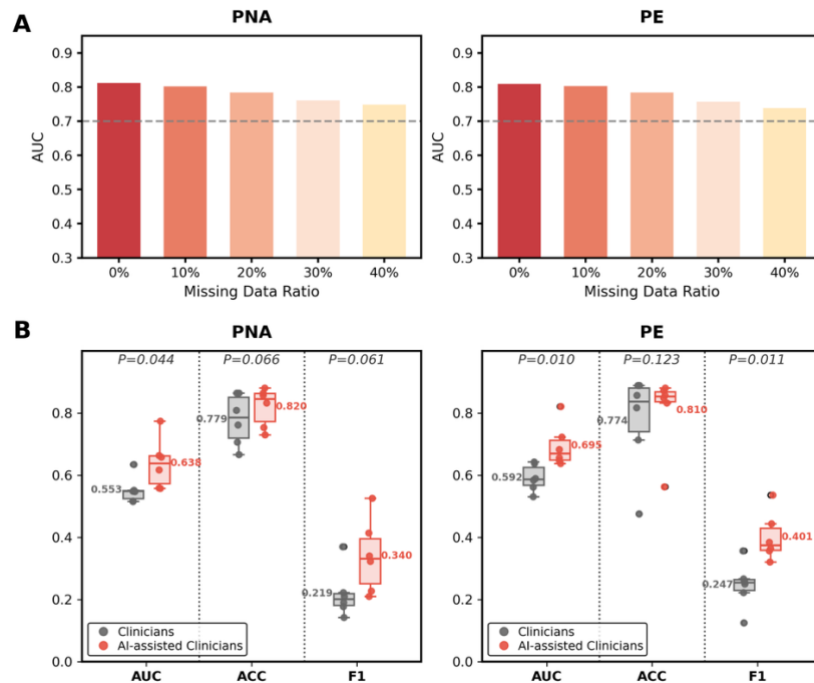


Fig. S2. Performance of our model as an AI assistant for clinicians. (A) Bar charts illustrating the model's AUC performance under simulated missing data ratios ranging from 0% to 40% for both PNA and PE. (B) Performance metrics for clinicians and model-assisted clinicians in the early detection of PNA and PE. Abbreviations: PNA, Pneumonia; PE, Pleural effusion.

SUPPLEMENTARY TABLE I

BASELINE CHARACTERISTICS IN THE DEVELOPMENT COHORT AND EVALUATION COHORT

Characteristic	Development cohort	Evaluation cohort	p value
Age	56 (47, 63)	55.5 (46, 61)	0.450
Male, n (%)	438 (88)	108 (86)	0.676
BMI	22.8 (20.8, 25.0)	23.3 (20.8, 24.8)	0.444
Hypertension, n (%)	84 (17)	20 (16)	0.908
Diabetes, n (%)	62 (12)	16 (13)	1.0
Cirrhosis, n (%)	311 (62)	74 (59)	0.540
Fatty liver, n (%)	38 (8)	14 (11)	0.273
Portal hypertension, n (%)	131 (26)	35 (28)	0.806
Ascites, n (%)	15 (3)	4 (3)	1.0
Preop Factors			
Tumor size, cm	5.08 ± 3.29	4.98 ± 3.39	0.761
Tumor number, n	1.14 ± 0.45	1.13 ± 0.40	0.739
ICGR15, %	5.1 (3.1, 7.6)	3.9 (2.7, 5.5)	0.008**
AFP, ng/mL	24.3 (4.2, 363.4)	20.9 (3.2, 248.9)	0.305
CEA, ng/mL	1.75 (0.94, 2.70)	1.80 (0.98, 2.59)	0.858
CA199, ng/mL	12.17 (6.21, 19.65)	13.97 (6.42, 19.40)	0.747
HBVs Ag(+), n (%)	424 (85)	110 (87)	0.570
HCV Ab(+), n (%)	16 (3)	5 (4)	0.590
Preop PT-INR	1.00 (0.95, 1.07)	1.01 (0.95, 1.06)	0.697
Preop WBC, 10 ⁹ /L	5.65 (4.63, 7.02)	5.72 (5.01, 7.02)	0.528
Preop LY, 10 ⁹ /L	1.66 (1.29, 2.08)	1.60 (1.31, 2.06)	0.669
Preop NE, 10 ⁹ /L	3.24 (2.47, 4.44)	3.44 (2.57, 4.17)	0.585
Preop RBC, 10 ¹² /L	4.56 (4.17, 4.97)	4.69 (4.33, 5.08)	0.054
Preop HGB, g/L	140 (129, 149)	136 (126, 148)	0.351
Preop PLT, 10 ⁹ /L	175.0 (128.5, 222.5)	182.5 (147.0, 221.5)	0.337
Preop ALT, U/L	28 (20, 39)	26 (17, 41)	0.479
Preop AST, U/L	28 (21, 39)	25.5 (20, 34)	0.088
Preop TP, g/L	67.5 (63.9, 71.4)	66.75 (63.05, 71.68)	0.482
Preop ALB, g/L	39.23 ± 3.93	39.36 ± 4.07	0.743
Preop TBIL, μmol/L	12.1 (9.5, 16.2)	13.1 (8.9, 15.8)	0.776
Preop Cr, μmol/L	77 (69, 87)	75 (67, 88)	0.405
Preop K, mmol/L	3.97 ± 0.35	3.95 ± 0.32	0.582
Preop Na, mmol/L	141 (139, 142)	140 (139, 142)	0.220
Intraop Factors			
Methods			0.106
Open	83 (17)	26 (21)	
Laparoscope	381 (76)	97 (77)	
Trans	35 (7)	3 (2)	
Major	75 (15)	28 (22)	0.070
ASA			0.549
1	30 (6)	9 (8)	
2	391 (84)	97 (86)	
3	42 (9)	7 (6)	
Duration of anesthesia, min	378.30 ± 128.08	359.71 ± 110.67	0.175
Duration of intubation, min	297.59 ± 115.50	287.94 ± 103.59	0.441
Crystalloid infusion, ml	1872.79 ± 775.73	1800.88 ± 758.00	0.373
Colloid infusion, ml	581.71 ± 405.03	552.63 ± 341.58	0.480
Total input, ml	2566.88 ± 1107.27	2470.22 ± 1014.71	0.397
Total output, ml	672.17 ± 587.69	635.18 ± 527.87	0.540

Operation time, min	204.91 ± 100.19	207.85 ± 95.41	0.766
Pringle, min	15.66 ± 23.14	17.14 ± 26.85	0.534
Bleeding, ml	214.29 ± 279.85	219.13 ± 393.97	0.874
Transfusion, ml	149.00 ± 348.04	140.48 ± 381.46	0.810
Postop Factors			
Postop 24h PT-INR	1.13 (1.05, 1.22)	1.13 (1.03, 1.23)	0.808
Postop 24h PCT, ng/ml	0.58 (0.25, 1.17)	0.46 (0.28, 1.03)	0.752
Postop 24h WBC, 10 ⁹ /L	12.35 (9.61, 15.52)	12.01 (9.87, 14.39)	0.417
Postop 24h LY, 10 ⁹ /L	0.74 (0.51, 1.03)	0.78 (0.55, 1.01)	0.371
Postop 24h NE, 10 ⁹ /L	10.55 (8.1, 13.43)	10.06 (8.51, 12.62)	0.550
Postop 24h RBC, 10 ¹² /L	4.23 (3.8, 4.59)	4.36 (3.82, 4.86)	0.041*
Postop 24h HGB, g/L	127.28 ± 17.37	128.22 ± 18.68	0.610
Postop 24h PLT, 10 ⁹ /L	144 (106, 184)	152 (116, 194.5)	0.217
Postop 24h ALT, U/L	174 (98.5, 317)	194.5 (106.8, 339.3)	0.446
Postop 24h AST, U/L	196 (109, 311)	186.5 (119.3, 311.8)	0.788
Postop 24h TP, g/L	58.54 ± 6.05	58.93 ± 6.46	0.545
Postop 24h ALB, g/L	34.6 (31.6, 37.4)	34.55 (32.12, 38)	0.405
Postop 24h TBIL, μmol/L	15.9 (10.7, 23.2)	16.45 (10.7, 25.42)	0.626
Postop 24h Cr, μmol/L	74 (67, 86)	71 (62, 83.75)	0.044*
Postop 24h K, mmol/L	4.12 (3.89, 4.42)	4.05 (3.86, 4.37)	0.172
Postop 24h Na, mmol/L	139 (137, 141)	139 (137, 141)	0.829

“*” indicates $0.01 < p \leq 0.05$, “***” indicates $0.001 < p \leq 0.01$. Abbreviations: AFP, α -fetoprotein; ALB, Albumin; ALT, Alanine aminotransferase; AST, Aspartate aminotransferase; BMI, Body mass index; CA199, Carbohydrate Antigen 19-9; CEA, Carcinoembryonic Antigen; Cr, Creatinine; HBV Ag Hepatitis B virus antigen; HCV Ab Hepatitis C virus antibody; HGB, Hemoglobin; ICG-R15, Indocyanine green retention rate at 15 minutes; K, Potassium; LY, Lymphocyte; Na, Sodium; NE, Neutrophil; PLT, Platelet; PT-INR, Prothrombin time-international normalized ratio; RBC, red blood cell; TBIL, Total bilirubin; TP, Total protein; WBC, White blood cell. ASA, American Society of Anesthesiologists Physical Status Classification System; Major, Major liver resection; PCT, Procalcitonin; Preop, Preoperation; Postop, Postoperation.

SUPPLEMENTARY TABLE II

ABLATION STUDY OF MULTI-TASK LEARNING ON CCI PREDICTION

Setting	CCI AUC(%)
STL	78.7±0.6
MTL	80.0±0.8
Remove PHLF	79.7±1.1
Remove PNA	77.9±1.4
Remove PE	79.5±0.8
Remove ASC	79.8±1.6

Abbreviations: STL, single-task learning; MTL, multi-task learning; CCI, Comprehensive Complication Index; PHLF, Post-hepatectomy liver failure; PNA, Pneumonia; PE, Pleural effusion; ASC, Ascites.